

Tasks for Package installation and Prompting Engineering

In this task, you'll learn how to install the package required in our project and how to write prompt to generate desirable results for large language models.

The task and example acquired and modified from:

1. Ahmed, Kamran. "Prompt Engineering Roadmap." Roadmap.Sh, roadmap.sh, 20 May 2023, roadmap.sh/prompt-engineering.
2. "ChatGPT Prompt Engineering for Developers" by DeepLearning.AI and OpenAI.

Setup

Load the API key and relevant Python libraries.

In this course, we've provided some code that loads the OpenAI API key for you.

```
In [1]: # install langchain in your environment
!pip install langchain
```

Collecting langchain
Downloading langchain-0.3.0-py3-none-any.whl.metadata (7.1 kB)
Requirement already satisfied: PyYAML>=5.3 in /usr/local/lib/python3.10/dist-packages (from langchain) (6.0.2)
Requirement already satisfied: SQLAlchemy<3,>=1.4 in /usr/local/lib/python3.10/dist-packages (from langchain) (2.0.34)
Requirement already satisfied: aiohttp<4.0.0,>=3.8.3 in /usr/local/lib/python3.10/dist-packages (from langchain) (3.10.5)
Requirement already satisfied: async-timeout<5.0.0,>=4.0.0 in /usr/local/lib/python3.10/dist-packages (from langchain) (4.0.3)
Collecting langchain-core<0.4.0,>=0.3.0 (from langchain)
Downloading langchain_core-0.3.1-py3-none-any.whl.metadata (6.2 kB)
Collecting langchain-text-splitters<0.4.0,>=0.3.0 (from langchain)
Downloading langchain_text_splitters-0.3.0-py3-none-any.whl.metadata (2.3 kB)
Collecting langsmith<0.2.0,>=0.1.17 (from langchain)
Downloading langsmith-0.1.122-py3-none-any.whl.metadata (13 kB)
Requirement already satisfied: numpy<2,>=1 in /usr/local/lib/python3.10/dist-packages (from langchain) (1.26.4)
Requirement already satisfied: pydantic<3.0.0,>=2.7.4 in /usr/local/lib/python3.10/dist-packages (from langchain) (2.9.1)
Requirement already satisfied: requests<3,>=2 in /usr/local/lib/python3.10/dist-packages (from langchain) (2.32.3)
Collecting tenacity!=8.4.0,<9.0.0,>=8.1.0 (from langchain)
Downloading tenacity-8.5.0-py3-none-any.whl.metadata (1.2 kB)
Requirement already satisfied: aiohappyeyeballs>=2.3.0 in /usr/local/lib/python3.10/dist-packages (from aiohttp<4.0.0,>=3.8.3->langchain) (2.4.0)
Requirement already satisfied: aiosignal>=1.1.2 in /usr/local/lib/python3.10/dist-packages (from aiohttp<4.0.0,>=3.8.3->langchain) (1.3.1)
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.10/dist-packages (from aiohttp<4.0.0,>=3.8.3->langchain) (24.2.0)
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.10/dist-packages (from aiohttp<4.0.0,>=3.8.3->langchain) (1.4.1)
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.10/dist-packages (from aiohttp<4.0.0,>=3.8.3->langchain) (6.1.0)
Requirement already satisfied: yarl<2.0,>=1.0 in /usr/local/lib/python3.10/dist-packages (from aiohttp<4.0.0,>=3.8.3->langchain) (1.11.1)
Collecting jsonpatch<2.0,>=1.33 (from langchain-core<0.4.0,>=0.3.0->langchain)
Downloading jsonpatch-1.33-py2.py3-none-any.whl.metadata (3.0 kB)
Requirement already satisfied: packaging<25,>=23.2 in /usr/local/lib/python3.10/dist-packages (from langchain-core<0.4.0,>=0.3.0->langchain) (24.1)
Requirement already satisfied: typing-extensions>=4.7 in /usr/local/lib/python3.10/dist-packages (from langchain-core<0.4.0,>=0.3.0->langchain) (4.12.2)
Collecting httpx<1,>=0.23.0 (from langsmith<0.2.0,>=0.1.17->langchain)
Downloading httpx-0.27.2-py3-none-any.whl.metadata (7.1 kB)
Collecting orjson<4.0.0,>=3.9.14 (from langsmith<0.2.0,>=0.1.17->langchain)
Downloading orjson-3.10.7-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (50 kB)

50.4/50.4 kB 1.5 MB/s eta 0:00:00

Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.10/dist-packages (from pydantic<3.0.0,>=2.7.4->langchain) (0.7.0)
Requirement already satisfied: pydantic-core==2.23.3 in /usr/local/lib/python3.10/dist-packages (from pydantic<3.0.0,>=2.7.4->langchain) (2.23.3)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.10/dist-packages (from requests<3,>=2->langchain) (3.3.2)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.10/dist-packages (from requests<3,>=2->langchain) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.10/dist-packages (from requests<3,>=2->langchain) (2.0.7)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.10/dist-packages (from requests<3,>=2->langchain) (2024.8.30)
Requirement already satisfied: greenlet!=0.4.17 in /usr/local/lib/python3.10/dist-packages (from SQLAlchemy<3,>=1.4->langchain) (3.1.0)

```

Requirement already satisfied: anyio in /usr/local/lib/python3.10/dist-packages (from httpx<1,>=0.23.0->langsmith<0.2.0,>=0.1.17->langchain) (3.7.1)
Collecting httpcore==1.* (from httpx<1,>=0.23.0->langsmith<0.2.0,>=0.1.17->langchain)
  Downloading httpcore-1.0.5-py3-none-any.whl.metadata (20 kB)
Requirement already satisfied: sniffio in /usr/local/lib/python3.10/dist-packages (from httpx<1,>=0.23.0->langsmith<0.2.0,>=0.1.17->langchain) (1.3.1)
Collecting h11<0.15,>=0.13 (from httpcore==1.*->httpx<1,>=0.23.0->langsmith<0.2.0,>=0.1.17->langchain)
  Downloading h11-0.14.0-py3-none-any.whl.metadata (8.2 kB)
Collecting jsonpointer>=1.9 (from jsonpatch<2.0,>=1.33->langchain-core<0.4.0,>=0.3.0->langchain)
  Downloading jsonpointer-3.0.0-py2.py3-none-any.whl.metadata (2.3 kB)
Requirement already satisfied: exceptiongroup in /usr/local/lib/python3.10/dist-packages (from anyio->httpx<1,>=0.23.0->langsmith<0.2.0,>=0.1.17->langchain) (1.2.2)
Downloading langchain-0.3.0-py3-none-any.whl (1.0 MB)
  ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.0/1.0 MB 23.3 MB/s eta 0:00:00
Downloading langchain_core-0.3.1-py3-none-any.whl (405 kB)
  ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 405.1/405.1 kB 22.8 MB/s eta 0:00:00
Downloading langchain_text_splitters-0.3.0-py3-none-any.whl (25 kB)
Downloading langsmith-0.1.122-py3-none-any.whl (289 kB)
  ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 290.0/290.0 kB 18.6 MB/s eta 0:00:00
Downloading tenacity-8.5.0-py3-none-any.whl (28 kB)
Downloading httpx-0.27.2-py3-none-any.whl (76 kB)
  ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 76.4/76.4 kB 5.0 MB/s eta 0:00:00
Downloading httpcore-1.0.5-py3-none-any.whl (77 kB)
  ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 77.9/77.9 kB 5.1 MB/s eta 0:00:00
Downloading jsonpatch-1.33-py2.py3-none-any.whl (12 kB)
Downloading orjson-3.10.7-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (141 kB)
  ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 141.9/141.9 kB 8.8 MB/s eta 0:00:00
Downloading jsonpointer-3.0.0-py2.py3-none-any.whl (7.6 kB)
Downloading h11-0.14.0-py3-none-any.whl (58 kB)
  ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 58.3/58.3 kB 3.8 MB/s eta 0:00:00
Installing collected packages: tenacity, orjson, jsonpointer, h11, jsonpatch, httpcore, httpx, langsmith, langchain-core, langchain-text-splitters, langchain
  Attempting uninstall: tenacity
    Found existing installation: tenacity 9.0.0
    Uninstalling tenacity-9.0.0:
      Successfully uninstalled tenacity-9.0.0
Successfully installed h11-0.14.0 httpcore-1.0.5 httpx-0.27.2 jsonpatch-1.33 jsonpointer-3.0.0 langchain-0.3.0 langchain-core-0.3.1 langchain-text-splitters-0.3.0 langsmith-0.1.122 orjson-3.10.7 tenacity-8.5.0

```

```

In [2]: # install openai in your environment
!pip install openai==0.28

```

Collecting openai==0.28

Downloading openai-0.28.0-py3-none-any.whl.metadata (13 kB)

Requirement already satisfied: requests>=2.20 in /usr/local/lib/python3.10/dist-packages (from openai==0.28) (2.32.3)

Requirement already satisfied: tqdm in /usr/local/lib/python3.10/dist-packages (from openai==0.28) (4.66.5)

Requirement already satisfied: aiohttp in /usr/local/lib/python3.10/dist-packages (from openai==0.28) (3.10.5)

Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.10/dist-packages (from requests>=2.20->openai==0.28) (3.3.2)

Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.10/dist-packages (from requests>=2.20->openai==0.28) (3.10)

Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.10/dist-packages (from requests>=2.20->openai==0.28) (2.0.7)

Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.10/dist-packages (from requests>=2.20->openai==0.28) (2024.8.30)

Requirement already satisfied: aiohappyeyeballs>=2.3.0 in /usr/local/lib/python3.10/dist-packages (from aiohttp->openai==0.28) (2.4.0)

Requirement already satisfied: aiosignal>=1.1.2 in /usr/local/lib/python3.10/dist-packages (from aiohttp->openai==0.28) (1.3.1)

Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.10/dist-packages (from aiohttp->openai==0.28) (24.2.0)

Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.10/dist-packages (from aiohttp->openai==0.28) (1.4.1)

Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.10/dist-packages (from aiohttp->openai==0.28) (6.1.0)

Requirement already satisfied: yarl<2.0,>=1.0 in /usr/local/lib/python3.10/dist-packages (from aiohttp->openai==0.28) (1.11.1)

Requirement already satisfied: async-timeout<5.0,>=4.0 in /usr/local/lib/python3.10/dist-packages (from aiohttp->openai==0.28) (4.0.3)

Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.10/dist-packages (from multidict<7.0,>=4.5->aiohttp->openai==0.28) (4.12.2)

Downloading openai-0.28.0-py3-none-any.whl (76 kB)

76.5/76.5 kB 1.7 MB/s eta 0:00:00

Installing collected packages: openai

Successfully installed openai-0.28.0

In [3]: `pip install -U langchain-community`

Collecting langchain-community

Downloading langchain_community-0.3.0-py3-none-any.whl.metadata (2.8 kB)

Requirement already satisfied: PyYAML>=5.3 in /usr/local/lib/python3.10/dist-packages (from langchain-community) (6.0.2)

Requirement already satisfied: SQLAlchemy<3,>=1.4 in /usr/local/lib/python3.10/dist-packages (from langchain-community) (2.0.34)

Requirement already satisfied: aiohttp<4.0.0,>=3.8.3 in /usr/local/lib/python3.10/dist-packages (from langchain-community) (3.10.5)

Collecting dataclasses-json<0.7,>=0.5.7 (from langchain-community)

Downloading dataclasses_json-0.6.7-py3-none-any.whl.metadata (25 kB)

Requirement already satisfied: langchain<0.4.0,>=0.3.0 in /usr/local/lib/python3.10/dist-packages (from langchain-community) (0.3.0)

Requirement already satisfied: langchain-core<0.4.0,>=0.3.0 in /usr/local/lib/python3.10/dist-packages (from langchain-community) (0.3.1)

Requirement already satisfied: langsmith<0.2.0,>=0.1.112 in /usr/local/lib/python3.10/dist-packages (from langchain-community) (0.1.122)

Requirement already satisfied: numpy<2,>=1 in /usr/local/lib/python3.10/dist-packages (from langchain-community) (1.26.4)

Collecting pydantic-settings<3.0.0,>=2.4.0 (from langchain-community)

Downloading pydantic_settings-2.5.2-py3-none-any.whl.metadata (3.5 kB)

Requirement already satisfied: requests<3,>=2 in /usr/local/lib/python3.10/dist-packages (from langchain-community) (2.32.3)

Requirement already satisfied: tenacity!=8.4.0,<9.0.0,>=8.1.0 in /usr/local/lib/python3.10/dist-packages (from langchain-community) (8.5.0)

Requirement already satisfied: aiohappyeyeballs>=2.3.0 in /usr/local/lib/python3.10/dist-packages (from aiohttp<4.0.0,>=3.8.3->langchain-community) (2.4.0)

Requirement already satisfied: aiosignal>=1.1.2 in /usr/local/lib/python3.10/dist-packages (from aiohttp<4.0.0,>=3.8.3->langchain-community) (1.3.1)

Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.10/dist-packages (from aiohttp<4.0.0,>=3.8.3->langchain-community) (24.2.0)

Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.10/dist-packages (from aiohttp<4.0.0,>=3.8.3->langchain-community) (1.4.1)

Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.10/dist-packages (from aiohttp<4.0.0,>=3.8.3->langchain-community) (6.1.0)

Requirement already satisfied: yarl<2.0,>=1.0 in /usr/local/lib/python3.10/dist-packages (from aiohttp<4.0.0,>=3.8.3->langchain-community) (1.11.1)

Requirement already satisfied: async-timeout<5.0,>=4.0 in /usr/local/lib/python3.10/dist-packages (from aiohttp<4.0.0,>=3.8.3->langchain-community) (4.0.3)

Collecting marshmallow<4.0.0,>=3.18.0 (from dataclasses-json<0.7,>=0.5.7->langchain-community)

Downloading marshmallow-3.22.0-py3-none-any.whl.metadata (7.2 kB)

Collecting typing-inspect<1,>=0.4.0 (from dataclasses-json<0.7,>=0.5.7->langchain-community)

Downloading typing_inspect-0.9.0-py3-none-any.whl.metadata (1.5 kB)

Requirement already satisfied: langchain-text-splitters<0.4.0,>=0.3.0 in /usr/local/lib/python3.10/dist-packages (from langchain<0.4.0,>=0.3.0->langchain-community) (0.3.0)

Requirement already satisfied: pydantic<3.0.0,>=2.7.4 in /usr/local/lib/python3.10/dist-packages (from langchain<0.4.0,>=0.3.0->langchain-community) (2.9.1)

Requirement already satisfied: jsonpatch<2.0,>=1.33 in /usr/local/lib/python3.10/dist-packages (from langchain-core<0.4.0,>=0.3.0->langchain-community) (1.33)

Requirement already satisfied: packaging<25,>=23.2 in /usr/local/lib/python3.10/dist-packages (from langchain-core<0.4.0,>=0.3.0->langchain-community) (24.1)

Requirement already satisfied: typing-extensions>=4.7 in /usr/local/lib/python3.10/dist-packages (from langchain-core<0.4.0,>=0.3.0->langchain-community) (4.12.2)

Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.10/dist-packages (from langsmith<0.2.0,>=0.1.112->langchain-community) (0.27.2)

Requirement already satisfied: orjson<4.0.0,>=3.9.14 in /usr/local/lib/python3.10/dist-packages (from langsmith<0.2.0,>=0.1.112->langchain-community) (3.10.7)

Collecting python-dotenv>=0.21.0 (from pydantic-settings<3.0.0,>=2.4.0->langchain-community)

Downloading python_dotenv-1.0.1-py3-none-any.whl.metadata (23 kB)

Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.10/dist

-packages (from requests<3,>=2->langchain-community) (3.3.2)
 Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.10/dist-packages (from requests<3,>=2->langchain-community) (3.10)
 Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.10/dist-packages (from requests<3,>=2->langchain-community) (2.0.7)
 Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.10/dist-packages (from requests<3,>=2->langchain-community) (2024.8.30)
 Requirement already satisfied: greenlet!=0.4.17 in /usr/local/lib/python3.10/dist-packages (from SQLAlchemy<3,>=1.4->langchain-community) (3.1.0)
 Requirement already satisfied: anyio in /usr/local/lib/python3.10/dist-packages (from httpx<1,>=0.23.0->langsmith<0.2.0,>=0.1.112->langchain-community) (3.7.1)
 Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.10/dist-packages (from httpx<1,>=0.23.0->langsmith<0.2.0,>=0.1.112->langchain-community) (1.0.5)
 Requirement already satisfied: sniffio in /usr/local/lib/python3.10/dist-packages (from httpx<1,>=0.23.0->langsmith<0.2.0,>=0.1.112->langchain-community) (1.3.1)
 Requirement already satisfied: h11<0.15,>=0.13 in /usr/local/lib/python3.10/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->langsmith<0.2.0,>=0.1.112->langchain-community) (0.14.0)
 Requirement already satisfied: jsonpointer>=1.9 in /usr/local/lib/python3.10/dist-packages (from jsonpatch<2.0,>=1.33->langchain-core<0.4.0,>=0.3.0->langchain-community) (3.0.0)
 Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.10/dist-packages (from pydantic<3.0.0,>=2.7.4->langchain<0.4.0,>=0.3.0->langchain-community) (0.7.0)
 Requirement already satisfied: pydantic-core==2.23.3 in /usr/local/lib/python3.10/dist-packages (from pydantic<3.0.0,>=2.7.4->langchain<0.4.0,>=0.3.0->langchain-community) (2.23.3)
 Collecting mpy-extensions>=0.3.0 (from typing-inspect<1,>=0.4.0->dataclasses-json<0.7,>=0.5.7->langchain-community)
 Downloading mpy_extensions-1.0.0-py3-none-any.whl.metadata (1.1 kB)
 Requirement already satisfied: exceptiongroup in /usr/local/lib/python3.10/dist-packages (from anyio->httpx<1,>=0.23.0->langsmith<0.2.0,>=0.1.112->langchain-community) (1.2.2)
 Downloading langchain_community-0.3.0-py3-none-any.whl (2.3 MB)
 2.3/2.3 MB 35.6 MB/s eta 0:00:00
 Downloading dataclasses_json-0.6.7-py3-none-any.whl (28 kB)
 Downloading pydantic_settings-2.5.2-py3-none-any.whl (26 kB)
 Downloading marshmallow-3.22.0-py3-none-any.whl (49 kB)
 49.3/49.3 kB 3.3 MB/s eta 0:00:00
 Downloading python_dotenv-1.0.1-py3-none-any.whl (19 kB)
 Downloading typing_inspect-0.9.0-py3-none-any.whl (8.8 kB)
 Downloading mpy_extensions-1.0.0-py3-none-any.whl (4.7 kB)
 Installing collected packages: python-dotenv, mpy-extensions, marshmallow, typing-inspect, pydantic-settings, dataclasses-json, langchain-community
 Successfully installed dataclasses-json-0.6.7 langchain-community-0.3.0 marshmallow-3.22.0 mpy-extensions-1.0.0 pydantic-settings-2.5.2 python-dotenv-1.0.1 typing-inspect-0.9.0

```

In [4]: import os
        # Add you OpenAI API key here
        os.environ["OPENAI_API_KEY"] = "sk-5MQXhz4a-m-FcuAwDKpPoxgPa6o8j1hJZvJuiTE3P6T3BlbkFJtZW
  
```

```

In [5]: from langchain.agents import tool
        from langchain.llms.openai import OpenAIChat
        from langchain.prompts import PromptTemplate
        from langchain.llms import OpenAI

        from langchain.chat_models import ChatOpenAI
        from langchain import PromptTemplate, LLMChain
  
```

```

In [6]: # initial our LLM
        chat = OpenAIChat(temperature=0.0) # temperature is the degree of randomness of the mode
  
```

```
<ipython-input-6-aeba6830b030>:2: LangChainDeprecationWarning: The class `OpenAIChat` was deprecated in LangChain 0.0.1 and will be removed in 1.0. An updated version of the class exists in the langchain-openai package and should be used instead. To use it run `pip install -U langchain-openai` and import as `from langchain_openai import ChatOpenAI`.
chat = OpenAIChat(temperature=0.0) # temperature is the degree of randomness of the model's output
/usr/local/lib/python3.10/dist-packages/langchain_community/llms/openai.py:1076: UserWarning: You are trying to use a chat model. This way of initializing it is no longer supported. Instead, please use: `from langchain_community.chat_models import ChatOpenAI`
warnings.warn(
```

Basic Example

```
In [8]: # Translate English to German
prompt=f"""
Translate the text delimited by single quotes from English to German.

'What is your name?'

"""
chat(prompt)
```

```
Out[8]: "'Wie ist dein Name?'"
```

```
In [10]: # You can also write specific task separately.
text=f"""
'How are you doing?'
"""
prompt=f"""
Translate the {text} from English to Spanish.
"""
chat(prompt)
```

```
Out[10]: '¿Cómo estás?'
```

Task 1. Role Prompting

1.1 Example

```
In [11]: prompt=f"""
As a traffic engineer, how would you approach the problem of traffic congestion to minimize
"""
print(chat(prompt))
```


There are several strategies that a traffic engineer can employ to minimize the impact of traffic congestion on transportation systems. Some of these strategies include:

1. Implementing Intelligent Transportation Systems (ITS): ITS technologies such as traffic signal coordination, dynamic message signs, and traffic monitoring systems can help optimize traffic flow and reduce congestion.
2. Improving infrastructure: Upgrading roads, bridges, and intersections to accommodate higher traffic volumes can help alleviate congestion. Adding lanes, improving road geometry, and implementing roundabouts can all help improve traffic flow.
3. Promoting alternative modes of transportation: Encouraging the use of public transportation, biking, walking, and carpooling can help reduce the number of vehicles on the road and alleviate congestion.
4. Implementing congestion pricing: Charging drivers a fee to use certain roads or lanes during peak hours can help reduce congestion by encouraging drivers to carpool, use public transportation, or travel at off-peak times.
5. Implementing traffic management strategies: Using variable speed limits, ramp metering, and reversible lanes can help manage traffic flow and reduce congestion.
6. Conducting traffic studies: Analyzing traffic patterns, volumes, and congestion hotspots can help identify areas where improvements can be made to reduce congestion.

By employing a combination of these strategies, traffic engineers can work to minimize the impact of traffic congestion on transportation systems and improve overall traffic flow and efficiency.

```
In [12]: prompt=f"""
As a driver, how would you approach the problem of traffic congestion to minimize its im
"""
print(chat(prompt))
```

1. Plan ahead: Before heading out, check traffic conditions and consider alternative routes to avoid congested areas.
2. Use public transportation: Utilize buses, trains, or other forms of public transportation to reduce the number of vehicles on the road.
3. Carpool: Share a ride with coworkers, friends, or neighbors to reduce the number of vehicles on the road.
4. Avoid peak hours: Try to schedule your trips during off-peak hours to avoid the busiest times on the road.
5. Practice defensive driving: Stay alert, follow traffic rules, and be courteous to other drivers to help maintain a smooth flow of traffic.
6. Stay informed: Keep up to date on traffic conditions through apps, websites, or radio updates to make informed decisions while on the road.
7. Consider alternative modes of transportation: Walking, biking, or using ride-sharing services can help reduce congestion and emissions on the road.
8. Support infrastructure improvements: Advocate for better public transportation, bike lanes, and other infrastructure improvements to help alleviate traffic congestion in the long term.

1.2 Task. As a student, you feel uncomfortable today and need to have some rest therefore you can not go to Professor J's class. Write an email to Professor J to explain your current situation.


```
In [13]: prompt=f"""
As a student, you feel uncomfortable today and need to have some rest therefore you can
"""
print(chat(prompt))
```

Subject: Unable to Attend Class Today

Dear Professor J,

I hope this email finds you well. I am writing to inform you that I am feeling unwell today and will not be able to attend your class. I have been experiencing discomfort and believe that I need to rest in order to recover.

I apologize for any inconvenience this may cause and I assure you that I will catch up on any missed work and assignments as soon as possible. If there are any important announcements or updates from today's class, I would greatly appreciate it if you could share them with me.

Thank you for your understanding and support. I look forward to returning to class as soon as I am feeling better.

Sincerely,
[Your Name]

In []:

Task 2 Generate Structured Output

Example

```
In [14]: prompt = f"""
Generate a list of three made-up book titles along with their authors and genres.
Provide them in JSON format with the following keys:
book_id, title, author, genre.
"""
print(chat(prompt))
```

```
[
  {
    "book_id": 1,
    "title": "The Midnight Garden",
    "author": "Elena Rivers",
    "genre": "Fantasy"
  },
  {
    "book_id": 2,
    "title": "Echoes of the Past",
    "author": "Nathan Black",
    "genre": "Mystery"
  },
  {
    "book_id": 3,
    "title": "Whispers in the Wind",
    "author": "Samantha Reed",
    "genre": "Romance"
  }
]
```

Task: we want to generate a list of five songs along with their Singer, Publish Date, Type in JSON format.

```
In [15]: prompt = f"""
Please generate a list of five songs along with their Singer, Publish Date, Type in JSON
"""
print(chat(prompt))

{
  "songs": [
    {
      "title": "Shape of You",
      "singer": "Ed Sheeran",
      "publish_date": "2017-01-06",
      "type": "Pop"
    },
    {
      "title": "Bohemian Rhapsody",
      "singer": "Queen",
      "publish_date": "1975-10-31",
      "type": "Rock"
    },
    {
      "title": "Someone Like You",
      "singer": "Adele",
      "publish_date": "2011-01-24",
      "type": "Pop"
    },
    {
      "title": "Hotel California",
      "singer": "Eagles",
      "publish_date": "1977-12-08",
      "type": "Rock"
    },
    {
      "title": "Uptown Funk",
      "singer": "Mark Ronson ft. Bruno Mars",
      "publish_date": "2014-11-10",
      "type": "Funk"
    }
  ]
}
```

Task 3. Few shot learning

3.1. Example of Few Shot learning

```
In [16]: text=f"""
- Turn left and Go straight

- Turn left twice

- Go straight and Turn left

- Turn right twice and Go straight third times

"""
```

```
prompt=f"""
Your task is to translate {text} to the answer in a consistent style based on the example below.

Turn left and Turn right – Turn left + Turn right

Turn right twice – (Turn right)*2

Go straight twice – (Go straight)*2
"""
```

```
In [17]: print(chat(prompt))
```

- Turn left and Go straight: Turn left + Go straight
- Turn left twice: (Turn left)*2
- Go straight and Turn left: Go straight + Turn left
- Turn right twice and Go straight third times: (Turn right)*2 + (Go straight)*3

3.2 Task

Jake is new to the comic books and super hero movies. Some day he find some characters (Doctor Strange, Doctor Denstiny, Blue Bettle, Storm, Iron Fist, Daredevil, Green Arrow, Black Canary, Raven, Doctor Doom) that he don't know which company create them. But he know some famous comic characters and where they are from: Superman-DC, Captain America - Marvel, Batman - DC, Wonder Woman- DC, Iron man - Marvel, Spiderman - Marvel. Based on the example provided, can you help Jake find out which company: Doctor Strange, Doctor Denstiny, Blue Bettle, Storm, Iron Fist, Daredevil, Green Arrow, Black Canary, Raven, Doctor Doom from using Few shot learning.

```
In [19]: text=f"""
        Doctor Strange, Doctor Denstiny, Blue Bettle, Storm, Iron Fist, Daredevil, Green Arrow, Black Canary, Raven, Doctor Doom
        """

prompt=f"""
Your task is to translate {text} to the answer in a consistent format as the below example.
Superman-DC, Captain America - Marvel, Batman - DC, Wonder Woman- DC, Iron man - Marvel
"""
```

```
In [20]: print(chat(prompt))
```

Doctor Strange – Marvel, Doctor Destiny – DC, Blue Beetle – DC, Storm – Marvel, Iron Fist – Marvel, Daredevil – Marvel, Green Arrow – DC, Black Canary – DC, Raven – DC, Doctor Doom – Marvel

Task 4. Chain of Thought

Example We want to combine the last letter of each word as our output

Without Chain of Thought

```
In [21]: prompt=f"""
Q:good things are
A:dse

Q:large Language Model
A:eel
```

```
Q:deep learning
A:pg

Q:autonomous vehicle
A: se

Q: turn left world
A:
""""
```

```
In [22]: print(chat(prompt))
```

peaceful and harmonious

With Chain of Thought

```
In [23]: prompt=f"""
Q:good things are
A:the last letter of 'good' is 'd'. the last letter of 'things' is 's'. the last letter

Q:large Language Model
A:the last letter of 'large' is 'e'. the last letter of 'Language' is 'e'. the last lett

Q:deep learning
A:the last letter of 'deep' is 'p'. the last letter of 'learning' is 'g'. so the output

Q:autonomous vehicle
A: the last letter of 'autonomous' is 's'. the last letter of 'vehicle' is 'e'. so the o

Q: turn left world
A:
""""
```

```
In [24]: print(chat(prompt))
```

the last letter of 'turn' is 'n'. the last letter of 'left' is 't'. the last letter of 'world' is 'd'. so the output of 'turn left world' is ntd

Task. We want to find odd numbers in number group: 15, 32, 5,13, 3 and see if the sum of those odd numbers is an even number. If the sum is even number, the answer is True, else the answer is False.

The example you may used in your Chain of Thought prompting is provided in the prompt without chain of thought below:

```
In [25]: # Without Chain of Thought
prompt=f"""
The odd numbers in this group add up to an even number: 4, 8, 9, 15, 12, 2, 1.
A: False.
The odd numbers in this group add up to an even number: 17, 10, 19, 4, 8, 12, 24.
A: True.
The odd numbers in this group add up to an even number: 16, 11, 14, 4, 8, 13, 24.
A: True.
The odd numbers in this group add up to an even number: 17, 9, 10, 12, 13, 4, 2.
A: False.
The odd numbers in this group add up to an even number: 15, 32, 5, 13, 3.
A:
""""
```

```
In [26]: print(chat(prompt))
```

False.

We can see without Chain of thought, the answer is wrong. Because the odd number in the group is 15, 5 and 13, 3, their sum is 36, which is an even number. Can you use Chain of Thought prompting to generate correct answer (You need to find the odd numbers in list and generate their sum, then check if their sum is correct or not)?

```
In [46]: prompt="""
The odd numbers in this group add up to an even number: 6, 10, 15, 21, 8, 4, 3.
A: Adding all the odd numbers (15, 21, 3) gives 39 so the answer is False.
The odd numbers in this group add up to an even number: 25, 14, 27, 6, 12, 18, 30.
A: Adding all the odd numbers (25, 27) gives 52 so the answer is True.
The odd numbers in this group add up to an even number: 20, 13, 16, 6, 10, 11, 28.
A: Adding all the odd numbers (13, 11) gives 24 so, the answer is True.
The odd numbers in this group add up to an even number: 23, 7, 8, 10, 11, 4, 2.
A: Adding all the odd numbers (23, 7, 11) gives 41, so the answer is False.
The odd numbers in this group add up to an even number: 17, 34, 9, 11, 78, 5, 1, 3, 2, 1
A:
"""
```

```
In [47]: print(chat(prompt))
```

Adding all the odd numbers (17, 9, 11, 5, 1, 3, 19, 21, 25, 79, 77, 93) gives 270, so the answer is True.

Below is a given example:

```
In [42]: prompt="""
The odd numbers in this group add up to an even number: 4, 8, 9, 15, 12, 2, 1.
A: Adding all the odd numbers (9, 15, 1) gives 25. The answer is False.
The odd numbers in this group add up to an even number: 17, 10, 19, 4, 8, 12, 24.
A: Adding all the odd numbers (17, 19) gives 36. The answer is True.
The odd numbers in this group add up to an even number: 16, 11, 14, 4, 8, 13, 24.
A: Adding all the odd numbers (11, 13) gives 24. The answer is True.
The odd numbers in this group add up to an even number: 17, 9, 10, 12, 13, 4, 2.
A: Adding all the odd numbers (17, 9, 13) gives 39. The answer is False.
The odd numbers in this group add up to an even number: 15, 32, 5, 13, 82, 7, 1, 3, 2, 1
A:
"""
```

```
In [43]: print(chat(prompt))
```

Adding all the odd numbers (15, 5, 13, 7, 1, 3, 17, 19, 23, 83, 73, 91) gives 260. The answer is True.

Task 5 Zero Shot Chain of Thought

Task. Jane have two different ways to commute to work: Option 1: She need to take bus first which takes her 1000 minutes, then she need to take a half hour train. Then She need a 10 minutes ride to work. Option 2: She need to take a 30 minute walk first, then she need to take 800 minute bus and an hour train. Finally she need a 30 minute bike ride. We can see it is a long journey. So can you use LLM to tell Jane which one is faster to get to work?

6.1 Just generate the prompt based on Zero Shot without Chain of thought

```
In [48]: prompt=f"""
Read the below options, and choose the option which is faster to get to work.

Option 1: She need to take bus first which takes her 1000 minutes, then she need to take
Option 2: She need to take a 30 minute walk first, then she need to take 800 minute bus

"""
```

```
In [49]: print(chat(prompt))
```

Option 1: She need to take bus first which takes her 1000 minutes, then she need to take a half hour train. Then She need a 10 minutes ride to work.

Is it a correct answer? You can also try to use Zero Shot Chain of Thought prompting to see if it give us the correct answer.

```
In [50]: prompt=f"""
Read the below options, and choose the option which is faster to get to work by converti

Option 1: She need to take bus first which takes her 1000 minutes, then she need to take
Option 2: She need to take a 30 minute walk first, then she need to take 800 minute bus

"""
```

```
In [51]: print(chat(prompt))
```

Let's convert the times to minutes:

Option 1:

- Bus: 1000 minutes
- Train: 30 minutes
- Ride: 10 minutes

Total time: $1000 + 30 + 10 = 1040$ minutes

Option 2:

- Walk: 30 minutes
- Bus: 800 minutes
- Train: 60 minutes
- Bike ride: 30 minutes

Total time: $30 + 800 + 60 + 30 = 920$ minutes

Therefore, Option 2 is faster with a total time of 920 minutes.

Collaboration

Please state who complete which task in next cell. If you complete all tasks together, then you can just enter : "We complete all tasks together"

```
In [52]: # Team member 1: We complete all tasks together
# Team member 2: We complete all tasks together
```

Model Limitations: Hallucinations (Not a task) You can see LLM produce fake view

- Boie is a real company, the product name is not real.

```
In [53]: prompt = f"""  
Tell me about AeroGlide UltraSlim Smart Toothbrush by Boie  
"""  
print(chat(prompt))
```

The AeroGlide UltraSlim Smart Toothbrush by Boie is a high-tech toothbrush designed to provide a superior cleaning experience. It features ultra-soft bristles that are gentle on the gums and teeth, while still effectively removing plaque and debris. The toothbrush also has a slim design that makes it easy to maneuver and reach all areas of the mouth.

One of the standout features of the AeroGlide UltraSlim Smart Toothbrush is its smart technology. It connects to a mobile app that tracks your brushing habits and provides personalized recommendations for improving your oral hygiene routine. The app also includes a timer to ensure you are brushing for the recommended two minutes.

Overall, the AeroGlide UltraSlim Smart Toothbrush is a sleek and innovative option for those looking to upgrade their oral care routine.

```
In [ ]:
```